Assignment No. 1.

Q1. Print an nxn star square
Take a dynamic input from the user & print an n x n square.
example:

If n=4:

```
* * * *
* * * *
* * * *
* * * *
```

→ Class Demo {
    PSVM () {
    int n=4;
for(int i=1; i<n; i++){
    for(int j=1; j<n; j++){
        Sop ("* ");
    }
    Sopln ();
    }
}
}

Dry run :-

int i=1

i<4

i<4 → j<4

int i=1 i 1 2 3 4 i++

i<4

1<4 → j<4 j++

i<4 → * 2

2<4 → * 3

3<4 → * 4

4<4 → * 5

$5 \leq 4$
↳ condition false.

$2 \leq 4$
↳ $1 \leq 4$ j++
↳ * 2

$2 \leq 4$
↳ * 3

$3 \leq 4$
↳ * 4

$4 \leq 4$ 5
↳ *

$5 \leq 4$
↳ false.

$3 \leq 4$
↳ $1 \leq 4$
↳ * 2

$2 \leq 4$
↳ * 3

$3 \leq 4$
↳ * 4

$4 \leq 4$
↳ * 5

$5 \leq 4$
↳ false.

$4 \leq 4$
↳ $1 \leq 4$
↳ * 2.

$2 \leq 4$
↳ * 3

$3 \leq 4$
↳ * 4

$4 \leq 4$
↳ * 5

$5 \leq 4$
↳ false.

$5 \leq 4$
↳ false.

O/p —

```
* * * *
* * * *
* * * *
* * * *
```

② Print following pattern

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
```

→

```
class Demo{
    psvm(){
        int n=4;
    for(int i=1; i<=n; i++){
        for(int j=1; j<=n; j++){
            sop(i+"-");
        }
        sopln();
    }
}
}
```

Dry run :

n = 4

int i = 1

1 ≤ 4

→ j ≤ n j++

1 ≤ 4

→ 1 2

2 ≤ 4

→ 1 3

3 ≤ 4

→ 1 4

4 ≤ 4

→ 1 5

5 ≤ 4

→ false.

i++

2 ≤ 4

→ 1 ≤ 4

→ 2 2

2 ≤ 4

→ 2 3

3 ≤ 4

→ 2 4

4 ≤ 4

→ 2 5

$3 \le 4$

$\rightarrow 1 \le 4$

$\rightarrow 3 \quad 2$

$2 \le 4$

$\rightarrow 3 \quad 3$

$3 \le 4$

$\rightarrow 3 \quad 4$

$4 \le 4$

$\rightarrow 3 \quad 5$

$5 \le 4$

$\rightarrow$ false

$4 \le 4$

$\rightarrow 1 \le 4$

$\rightarrow 4 \quad 2$

$2 \le 4$

$\rightarrow 4 \quad 3$

$3 \le 4$

$\rightarrow 4 \quad 4$

$4 \le 4$

$\rightarrow 4 \quad 5$

$5 \le 4$

$\rightarrow$ false

$5 \le 4$

$\rightarrow$ false.

O/P. –

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

Q.3) Print the following pattern

int n=4

1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4

```
class Demo {
    psvm ( ) {
        int n=4;
    for(int i=1; i≤n; i++){
        for( int j=1; j≤n; j++){
            sop ( j + "_" );
        }
        sopln ( );
    }
}
```

Dry run :

int n=4

i=1

1≤4 → j≤n j++

1≤4 → 1 2

2≤4 → 2 3

3≤4 → 3 4

4≤4 → 4 5

5≤4 → false.

i++

2≤4 → 1≤4 → 1 2 j++

2≤4 → 2 3

3≤4 → 3 4

4≤4 → 4 5

5 ≤4 → false

$3 \leq 4$ j++

$\rightarrow 1 \leq 4 \rightarrow 1 \quad 2$

$2 \leq 4 \rightarrow 2 \quad 3$

$3 \leq 4 \rightarrow 3 \quad 4$

$4 \leq 4 \rightarrow 4 \quad 5$

$5 \leq 4 \rightarrow$ false

$4 \leq 4$

$\rightarrow 1 \leq 4 \rightarrow 1 \quad 2$

$2 \leq 4 \rightarrow 2 \quad 3$

$3 \leq 4 \rightarrow 3 \quad 4$

$4 \leq 4 \rightarrow 4 \quad 5$

$5 \leq 4 \rightarrow$ false

$5 \leq 4$

$\rightarrow$ false.

O/P

1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4

Rows - outer loop
Columne:- Inner loop